

HW2, Due back to mycourses: 11:59pm, Tuesday, 2/3

**Design of Experiments
Worksheet 7 – Formula Derivation**

A. $Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, i = 1, \dots, a, j = 1, \dots, n, \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$. Show that

$$Y_{ij} \sim N(\mu + \tau_i, \sigma_\varepsilon^2)$$

$$\bar{Y}_{i.} \sim N(\mu + \tau_i, \frac{\sigma_\varepsilon^2}{n})$$

$$\bar{Y}_{..} \sim N(\mu, \frac{\sigma_\varepsilon^2}{an})$$

B. Recall that $s_Y^2 = \frac{\sum_i \sum_j (Y_{ij} - \bar{Y}_{..})^2}{N - 1}$ and $s_i^2 = \frac{\sum_j (Y_{ij} - \bar{Y}_{i.})^2}{n - 1}$

Based on the results from part A, give two unbiased estimators of σ_ε^2

C. We know that $SS_{Total} = (N - 1)s_Y^2$ and $MS_{Total} = \frac{SS_{Total}}{N - 1}$

$$SS_{Error} = \sum_{i=1}^a (n - 1)s_i^2 \text{ and } MS_{Error} = \frac{SS_{Error}}{N - a}$$

Show that $E(MS_{Total}) = \sigma_\varepsilon^2$ and $E(MS_{Error}) = \sigma_\varepsilon^2$

D. What do you conclude about MS_{Total} and MS_{Error} as an estimator of σ_ε^2 ?

E. Show that (choose one to work on)

$$SS_{Total} = \sum_{i=1}^a \sum_{j=1}^n Y_{ij}^2 - \frac{Y_{..}^2}{N}$$

$$SS_{Treatment} = n \left(\sum_{i=1}^a \bar{Y}_{i.}^2 - a \bar{Y}_{..}^2 \right) = \frac{1}{n} \sum_{i=1}^a Y_{i.}^2 - \frac{Y_{..}^2}{N}$$

$$SS_{Error} = \sum_{i=1}^a \sum_{j=1}^n Y_{ij}^2 - \frac{1}{n} \sum_{i=1}^a Y_{i.}^2$$